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MILITARY SPECIFICATION

GRINDING MACHINE, CENTER HOLE, VERTICAL, FLOOR MOUNTED, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electric motor driven, floor mounted, vertical spindle center hole grinding machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size	Workpiece length (Maximum) (Inches)	Center hole diameter range (Inches)
1	6	0 to 0.375
2	39	0 to 1.5
3	39	0.08 to 4.3
4	39	0.04 to 6
5	39	0.02 to 12
6	42	0.05 to 2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA) (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NFPA 79	- Electrical Standards for Industrial Machinery.
ANSI/NFPA 70	- National Electrical Code.
ANSI B7.1	- Safety Requirements for the Use, Care and Protection of Abrasive Wheels.
ANSI B11.9	- Safety Requirements for the Construction, Care, and Use of Grinding Machines.
ANSI B74.2	- Specifications for Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.
ANSI B74.13	- Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing.
AGMA 390.03	- Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or military standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current, commercial models. Each machine shall be capable of producing geometrically correct and precisely located center holes in work parts to tolerances specified herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repairs.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement

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system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station except as otherwise specified (see 6.2.1).

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the grinding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 29 CFR 1910 that are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.9 and OSHA 29 CFR 1910, the requirements of OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall

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have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Machine ways. Machine guideways shall be adequately protected whenever possible to exclude grinding dust, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members.

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The machine ways shall be hardened and ground and suitable for the high grade precision machines covered by this specification. Adjustable gibs or other means shall be provided to compensate for guideway wear.

3.3.9 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each machine shall consist primarily of a base, column, grinding head assembly, tailstock, vise, wheel dresser, driving mechanisms, and controls.

3.4.1 Base. The base shall be ribbed and braced to withstand any forces encountered during maximum operating conditions. The base shall have a machined support pad arranged and constructed to accept the column. Means shall be provided for floor mounting and leveling.

3.4.2 Column. The column shall be of box-like construction, ribbed and braced to minimize distortion and deflection under normal grinding loads. The column shall be arranged to support and carry the grinding head assembly, tailstock, workholding device, and controls. Guideways shall be provided to support vertical and horizontal movement of the head assembly, tailstock, and workholding device, as applicable. The column guideways shall be machined so the tailstock center and workholding device shall be, at any vertical adjustment on the column, on centerline with the wheelhead spindle. The column bottom surfaces shall be accurately fitted and rigidly secured to the mating surfaces of the base.

3.4.3 Grinding head assembly. The grinding head assembly shall support and carry the wheelhead and its drive mechanism, a system to provide reciprocating and, when applicable (see 3.4.4), planetary motion to the grinding wheel, and a grinding wheel dresser. The head assembly shall be mounted on the column guideways and be arranged to travel vertically or horizontally, as applicable. Head assembly movement shall be either manually or power actuated, as specified (see 6.2.1), with provisions for both rapid and fine adjustment. The head assembly shall be provided with a counterbalance.

3.4.3.1 Wheelhead. The wheelhead shall be fitted to the grinding head assembly in a vertical plane. The wheelhead spindle shall rotate in the wheelhead on bearings and be constructed to accept a driving mechanism at one end and a grinding wheel at the other end. The spindle and its related

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components shall be capable of vertical reciprocation and rotating at speeds in accordance with table I. Automatic mist lubrication shall be provided to the wheelhead. Air piping to the mist lubricator shall be provided with suitable filters to supply clean dry air for the system.

3.4.3.2 Wheelhead drive mechanism. The wheelhead drive mechanism shall be either an electric motor driven spindle with infinitely variable speed or a belt and pulley arrangement to provide for not less than three spindle speeds or a stepless, adjustable pneumatic spindle. Motor horsepowers shall be in accordance with table I. Means for adjusting belt tension shall be provided, as applicable, unless a nontensioning type belt is used.

3.4.3.3 Pivoting headstock center. When specified (see 6.2.1), a pivoting headstock center shall be provided to position noncylindrical workparts from existing center holes. The pivoting headstock center shall be mounted to the grinding head assembly in such a manner that when it is pivoted downward and comes to rest against its stop, the center is accurately positioned concentric with the axis of the planetary circle of the machine. The headstock center shall be provided with a handle to move it manually from its rest position. The center shall be held in place after positioning by a spring.

3.4.3.4 Grinding wheel dresser. The wheel dresser shall be of the diamond point type mounted on the grinding head assembly with a means for moving into position for dressing the grinding wheel. The wheel dresser shall provide a means for the diamond to travel the length of the wheel face while simultaneously oscillating across the wheel to produce a precise conical shape on the wheel regardless of the setting of the diamond in its shank or of the diamond wear. Initiation of the dressing cycle shall be operator controlled.

3.4.4 Planetary drive. Planetary motion of the grinding wheel or the workpiece, as specified (see 6.2.1), shall be provided. The planetary drive shall be powered by an electric motor and shall include an arrangement to provide planetary motion to the work axis of the grinding wheel or workpiece. The planetary drive mechanism shall be adjustable to supply a planetary circle diameter at bottom of stroke, the range of which is from not more than 0.005 inch to not less than 0.100 inch. Motor horsepower and spindle speed shall be in accordance with table I. Means for adjusting belt tension shall be provided unless a nontensioning belt is used.

3.4.5 Workholding device. The workholding device shall be mounted on the front of the column. Means shall be provided to adjust the workholding device horizontally or vertically, as applicable, with provisions for locking after adjustment. Horizontal or vertical adjustment shall be accomplished by movement of the workholding device or the grinding head assembly and shall allow any cylindrical object held in the workholding device to be aligned concentric with the axis of the wheelhead spindle. Covers or

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shields shall be provided to protect the guideways and sliding surfaces from contaminants.

3.4.6 Tailstock. The tailstock shall be mounted to the lower end of the column, be vertically adjustable, with means for clamping after adjustment. A handwheel shall be provided to manually actuate vertical movement. The tailstock shall be arranged on the column so the tailstock center is on the centerline with the workholding device and the wheelhead spindle. The tailstock shall be constructed to accept a center(s) and other workholding attachments. When specified (see 6.2.1), a tailstock work drive motor and a rotating work-holding arrangement shall be provided to allow the workpiece to be powered rotated. Motor horsepower and tailstock spindle speed shall be in accordance with table I.

3.4.7 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. Grinding wheels shall be properly guarded.

3.4.8 Electric system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.8.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements of dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG 1 and ANSI/NFPA 70. Pneumatic motors shall be of the vane or impulse reaction turbine air motor type controlled by the regulation of air pressure (see 3.4.10).

3.4.8.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.9 Coolant system. When specified (see 6.2.1), the machine shall be supplied with a coolant system. The system shall have adequate capacity for all types of grinding operations for which the machine is designed and shall have means for regulating the flow of and for adjusting spray coolant supply as required. All components necessary to make the system completely functional shall be furnished including reservoirs, screens, filters, drain troughs, and clean-out plates.

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3.4.10 Pneumatic system. When specified (see 6.2.1), a pneumatic system shall be provided. Shop air shall be provided having a line pressure of 90 to 120 psi. The machine shall have an air line filter, water trap, and oil mist lubricator. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. Each machine shall be capable of grinding center holes in workpieces in accordance with the requirements herein, and specifically in accordance with the test requirements designated in 4.7.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of 4.7.3.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special hand tools normally furnished by the manufacturer.

b. Aligning equipment necessary for aligning tailstock center, workholding device and headstock center.

c. Two each: 9/16, 1-1/8 and 2-inch diameter cone shaped grinding wheels, with 3/8-16 screw mountings.

3.9 Optional equipment. Optional equipment as described shall be furnished with each machine, when specified (see 6.2.1).

a. One adjustable worklight.

b. One vacuum dust collector with dust collector head assembly.

c. Special fixtures which control from the pitch diameter to hold spur or bevel gears.

d. One each: long stationary center, short stationary center, and longshanked, 3-1/2 inch diameter live center for use in tailstock.

e. One steadyrest for workparts 1-1/8 inch diameter to 9-inch diameter, including jaws to cover range.

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f. One 4-3/4 inch capacity chuck and adapter.

g. Magnetic chuck and adapter.

h. Chuck spindle assembly for holding chucks.

i. One extra set planetary drive belts.

j. One extra set wheelhead drive belts.

k. One holding device and indicator for repositioning centers and holding irregular shaped parts.

l. Additional optional equipment as required and specified by procuring activity.

3.10 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identify.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.12 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be fitted with a standard hour meter to measure operating time of the main drive motor. When specified (see 6.2.1), in addition to the meter on the main drive motor, the machine shall be fitted with a second hour meter to measure "spindle operating time." All meters shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.13 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the machine shall be commensurate will all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under normal environmental conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The machine and related equipment shall be examined for compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.14 including compliance with all referenced standards.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. Each machine shall be tested by performing the following tests. The instruments, test material and methods used in performing the tests shall be at the option of the supplier, provided they accurately reflect the values indicated. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1).

4.7.2.1 Angle test. Using a test bar located on the tailstock center and in the workholding device, finish grind a 60° center with a long spark out. Using two size balls, one 0.6250 inch in diameter and the other 0.3125 inch in diameter, measure the difference in height between the two balls and compute the angle of the center. The angle computed should be 60° to within +0°, -0° 5'.

4.7.2.2 Location test. Using a test bar 1 inch in diameter at one end and 5 inches in diameter at the other, locate the test bar from tailstock and workholding device. With the workholding device locating the test bar from the 5 inches in diameter end, grind the test bar. Remove the test bar, turn it, and locate the workholding device from the 1 inch diameter end.

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Grind the smaller end. Check the concentricity of each diameter with the test bar located on centers. The concentricity of each diameter should be within 0.001 inch TIR.

4.7.2.3 Roundness test. Grind a center in a test bar. Check the roundness of the center hole midway of the cone element. The center should be round to within ±0.00015 inch.

4.7.3 Alignment and accuracy tests. Unless otherwise specified (see 6.2.1), each machine and its components shall be tested for conformance with the accuracies specified in table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.10.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any shop for the preparation of workpieces for grinders that require the work to be held between centers and precision tolerance capabilities are needed.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Controls, if different (see 3.2.4).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.10).

j. Manually or power actuated head assembly movement (see 3.4.3).

k. Pivoting headstock center, if required (see 3.4.3.3).

l. Planetary motion of the grinding wheel or workpiece, as specified (see 3.4.4).

m. Tailstock work drive motor and rotating work-holding arrangement, if required (see 3.4.6).

n. Electrical system and input power, if different (see 3.4.8).

o. Motor enclosure, if different (see 3.4.8.1).

p. Coolant system, if required (see 3.4.9).

q. Pneumatic system, if required (see 3.4.10).

r. Size and capacity, if different (see 3.5).

s. Standard equipment, if different (see 3.8).

t. Specify and fully describe optional equipment, if required (see 3.9).

u. Specify applicable repair parts, if required (see 3.10).

v. Lubrication plate, if different (see 3.11.1).

w. Nameplate, if different (see 3.11.2).

x. Hour meter, if not required; "spindle operating time" meter, if required (see 3.12).

y. Inspection conditions, if different (see 4.3).

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z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. Performance tests, if different (see 4.7.2).

cc. Alignment and accuracy requirements, if different (see 4.7.3).

dd. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machine in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guards supplied to the commercial market. The procuring activity should specify size and configuration of workpieces for any required special point-of-operation guarding. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows: The basic machine covered in MIL-G-80179 corresponds to machine Size 6 in MIL-G-80179A. In addition to the one size from MIL-G-80179, five new machine sizes, Sizes 1 through 5, have been included in MIL-G-80179A.

6.6 Subject term (key word) listing.

Abrasive wheel
Center hole grinder

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Dresser, wheel
Grinder, center hole, vertical
Machine tool, grinding, center hole
Wheel, abrasive, grinding
Wheelhead, vertical spindle

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristic	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6
Workpiece:						
Workpiece length, maximum, inches	6	39	39	39	39	42
Workpiece length, maximum, extended column, inches	N/A	51	59/78	59/78	49/59/118	75/110
Workpiece length using workdrive, maximum, inches	N/A	N/A	39/59/78	39/59/78	39/49/59/118	38/71/106
Workpiece diameter, maximum, inches	0.375	1.5				
- using a steady rest	N/A	N/A	5	7.87	7.87	9
- using a chuck or workhead	N/A	N/A	N/A	7.87	4.72	6
Center:						
Center hole diameter, range, inches	0 to 0.375	0 to 1.5	0.08 to 4.3	0.04 to 6	0.02 to 12	0.05 to 2
Center hole taper angle, degrees	60	60 to 90	60 to 90	60 to 90	60 to 90	60
Wheelhead spindle:						
Grinding stroke (reciprocation), inches	1.5	1.187	10.5	2	0.39	2
Speed ranges, RPM:						
Workpiece drive	N/A	N/A	50 to 100	50 to 100	0 to 33	0 to 48
Wheelhead spindle	10,000 to 28,000	1,250 to 10,000	15,000 to 45,000	7,500 to 20,000	0 to 60,000	13,000 to 45,000
Planetary drive	N/A	N/A	10 to 100	N/A	N/A	0 to 330
Motor, HP:						
Wheelhead spindle motor	0.4	0.4	0.4	0.4	0.4	0.5
Workpiece drive motor	N/A	N/A	0.2	0.2	0.2	0.25
Planetary drive motor	N/A	N/A	0.2	N/A	N/A	0.25

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and accuracy requirements.

Component characteristics	Maximum allowable deviation
Head assembly to column, alignment	0.0005"/6"
Vise to column alignment	0.0008"
Tailstock center to planetary head alignment	0.0002" TIR
Planetary head square with column	0.0005"/6"
Headstock center with planetary head alignment	0.0002" TIR

Custodians:
Army - AL
Navy - None
Air Force - 99

Preparing activity:
DLA - IP

Project (3415-0158)

Review activities:
Army - AL
Navy - None
Air Force - 84
DLA - GS

User activities:
Army - AR, SM
Navy - AS, MC
Air Force - None

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80179A	Grinding Machine, Center Hole, Vertical, Floor Mounted, Electric Motor Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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